Scholarly Paper

Ontology-Based Approach to Risk-Factor Identification to Support the Management of Provisions in Bridge Design

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Abstract: Requirement management is crucial in the design stage of infrastructure project delivery to ensure the designed facility satisfies the owner expectations and avoids any costly redesign, rework, and disputes. The requirements are primarily described in design codes and standards. The relevant requirements from codes are transformed into a set of in computer-readable rules required for computer-aided design and compliance verification. Prior to the construction of rules, the requirements in codes must be converted into an organized, structured format in terms of risks to support easy retrieval of the relevant applicable requirements. However, the unavailability of risk information in the requirement text makes it challenging for a designer to extract the relevant requirements of specific risks to support design tasks, including compliance checking. The manual process of requirement classification according to risk factors is time-consuming, laborious, and error-prone because the codes are mostly voluminous, including thousands of specifications. Much less attention has been paid to develop an automated framework to organize the requirements in terms of risks addressed in them. To address this need, this study has attempted to develop an ontology-based framework to identify relevant risk factors addressed in bridge design requirements to support requirement management. The nine risk factors of bridge engineering used in this study included flood, earthquake, fire, snowfall, wind, temperature, overloading, vessel collision, and blast loading. A risk ontology was developed to represent the conceptualized semantic knowledge associated with each of the nine risk factors. The algorithm was validated on a human-annotated requirement dataset based on the AASHTO bridge design specifications and state design manuals. The developed model yielded a Spearman, Kendall tau, and Pearson correlation coefficient of 0.7223, 0.6021, and 0.7222, respectively. The proposed model is expected to improve specifications retrieval for rules construction, thereby enabling ease of requirement classification for the design process. **DOI: 10.1061/(ASCE)LA.1943-4170.0000574.** © *2022 American Society of Civil Engineers.*

Author keywords: Ontology; Text classification; Bridge design; Design codes; Requirements; Deep learning; Natural language processing; Text mining.

Introduction

Efficient requirement management is crucial in the design stage to ensure the designed infrastructure facility (i.e., bridges) conforms to all applicable requirements mentioned in different design codes and standards, thereby avoiding the redesign, rework, and disputes in later stages (Pritchard 2013). The requirements in codes are constructed to guarantee the facility can withstand the impact of all environmental hazards or risks such as flood, earthquake, and snowfall. (Björnsson 2017; Bulleit et al. 2015; Shapiro 1997). The understanding of risk factors of requirements is required for the efficient management of requirements and converting them into computer-readable rules for automated design and code verification to reduce legal issues due to the use of manual methods (Pohl 2020). The quality of design depends on how precisely the designer selects the applicable requirements of each risk factor from codes to convert them into rules (Al-dosary 2015). However, the civil infrastructure codes are not organized in terms of risk factors. In addition, the unavailability of risk information or objectives within the requirement text (e.g., in prescriptive requirements) makes it challenging for the designer to extract and organize the applicable requirements in terms of risk factors (flood, earthquake, etc.) addressed in them (Patil and Molenaar 2011; Hooton and Bickley 2012; Goodhead et al. 2016). The omission of applicable requirements of specific risk factors by young engineers have been reported as most common cause behind the structural failures, costly disputes, delays, financial loss, and increased project duration (Björnsson 2016, 2017). One of the best examples is the Nicoll Highway collapse in 2014 due to the imprecise selection of requirements for design (Palaneeswaran et al. 2014). Therefore, an adequate approach is needed that can extract and organize the applicable requirements based on risk factors to check conformance of design to each risk factor.

The precise extraction and management of applicable regulatory requirements is crucial to ensure the design can resist against different natural risks and help construction project management avoid undesirable legal issues (i.e., disputes and costly delays) (Bulleit 2012). However, transportation infrastructure (i.e., bridges) design codes are typically voluminous, including a large set of requirements associated with different risk factors. In addition, the requirements are often highly complex and vary a lot in terms of formatting, semantics, and the information they provide (Zhang and El-Gohary 2016). Moreover, the majority of requirements in codes are prescriptive in nature, providing a specific safety measure without explicitly specifying what type of risk being controlled (Goodhead et al. 2016; Hooton and Bickley 2012; Patil and Molenaar 2011). For instance, the following prescriptive provision, "The spacing of intermediate bracing shall be based on lateral stability and load transfer requirements but shall not exceed 25.0 ft" prescribes how the structure should be designed but does not

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explain the objective of provision, which is mitigation of the earthquake hazard. This writing style of design codes makes it challenging for engineers, especially those with little knowledge like young engineers and engineering students, to understand risk factors of requirements and to manage them in terms of risks (Lobo et al. 2006). Young engineers often blindly extract the requirements and assign possible risk categories without knowing their precise objectives (Coeckelbergh 2006). Therefore, the manual process of extracting the requirements from codes may result in situations where a few unusual requirements associated with a specific risk category may likely be missed in the design verification (Björnsson 2016). In such situations, the design verification for the risk scenarios outside the scope of the selected requirements is entirely disregarded, which may affect the performance of the bridge and result in legal issues (Björnsson 2016). Hence, the current state-of-the-art needs an efficient method that can support the extraction and management of precise set of requirements for bridge design verification of different environmental risks to ensure a safe and durable bridge design.

Currently, practitioners rely on the manual process of risk factor determination to extract and manage different types of relevant requirements for design and verification. This manual process is extremely time-consuming, tedious, and error prone. Although a few researchers have attempted to develop classification models to manage legal documents, no model supports the management of design codes in terms of environmental risks. The previously developed models supported the contract management tasks and the compliance checking of designs for a limited type of requirements only. To the best of authors' knowledge, no such related study or model is currently available that can predict the ranked list of relevant risk factors for bridge design requirements (prescriptive as well as performance-based) to enable extraction of a precise set of requirements for design verification. This study has attempted to fill the gap by developing an automated model that computes the semantic similarity between a requirement text and the risk ontologies to predict a ranked list of relevant risks addressed in the input requirements specifically for bridge engineering. To achieve this goal, the study leverages natural language processing (NLP) methods, including vector space models along with the use of domain ontologies to uncover the meanings of the technical terms by evaluating their context words in the design codes and standards. The study considers nine major types of risks experienced in the bridge projects for this domain-specific problem. These risks include flood, earthquake, snowfall, wind, vessel collision, temperature, blast loading, fire, and overloading. A domain specific ontology is developed for each risk category. Using the developed ontologies and the vector space model, the relevant risks addressed in a requirement are extracted and ranked according to the semantic similarity of the requirement text and the nine risk factors. The study is expected to help practitioners in managing the code provisions for the design and verification of bridge designs. The model can assist bridge engineers in correctly selecting relevant requirements for each potential risk factor to support requirement classification and management, which will eventually help improve design quality and reduce bridge failure.

Background

Role of Requirement Management in the Design of Bridges

The design stage of civil infrastructure systems, such as bridges, involves the management and organization of requirements in a structured format to enable a computer-aided design and verification, which eventually saves time, money, and avoids costly disputes at later stages due to imprecise design (Firth 2007). The responsibility of a bridge designer is to provide a cost-effective solution that meets the required performance and conforms to all minimum code standards of different risk factors to prevent any legal issue or a delay (Pritchard 2013). To achieve this objective, the challenge for a designer is to identify, evaluate, and verify the implemented requirements to control different risks involved in a specific project. For instance, the bridges located in seismic areas must conform to every major and minor seismic requirement mentioned in the codes (Madhusudhanan and Sideris 2018). For this, the designer is often required to manually extract the seismic related requirements from codes to check the conformance of bridge design to them. Missing any relevant requirements can lead to imprecise design and disputes. The design codes adopted to control the risks provide two types of requirements: (1) prescriptive requirements, and (2) performance-based requirements (Coeckelbergh 2006; Elms 1999; Hurd 2012). Prescriptive requirements demonstrate the acceptable specific solution to prevent a hazard, while the performance-based requirements illustrate the required performance against a risk. Among these, prescriptive provisions to control the risks in bridge designs can be further classified into the following two categories. The first type includes the provisions where the clear quantitative prescription is given. An example for this type of requirement is "The spacing of intermediate bracing shall be based on lateral stability and load transfer requirements but shall not exceed 25.0 ft." Another type of prescriptive requirements uses the references to other relevant codes, for example "End anchorages (nuts washers and anchor plates) shall be compatible with the tie rod system and shall be galvanized in accordance with AASHTO M 111." The extraction of prescriptive and performance-based requirements associated with different risk factors from codes is a challenging task. Young designers often experience confusion regarding the identification of the list of relevant risks addressed in prescriptive and performance-based requirements (Björnsson 2016). The missing of the requirements of any particular risk in design verification put the safety of the bridge at risk, resulting in costly litigation. Therefore, accurate extraction and management of the applicable requirements is highly important for design verification of a bridge project to produce a design that completely satisfies the wishes of the client.

Ontologies

Ontologies are a popular knowledge representation method that explicitly specifies the domain knowledge shared among many systems (Qin et al. 2018). They are widely employed as domain knowledge in several text-mining tasks such as text classification, text clustering, and word sense disambiguation (Allahyari et al. 2017). Ontologies typically represent the domain-specific knowledge in a machine-readable format specifically in the form of concepts and the relationships among them (El-Gohary and El-Diraby 2010; Keet 2004). Such a representation of semantic knowledge in ontologies allows knowledge sharing and reuse to develop many other applications in the future (Chang and Huang 2008). An ontology is composed of (1) classes indicating concepts in the domain of interest, and (2) relationships showing the semantic relationship between different concepts (Deviyanti et al. 2019).

Vector Space Model

Vector space models (VSM), commonly known as word embeddings, are an efficient approach to transforming textual data such as words, phrases, or sentences into numerical vectors (Erk 2012). VSM is developed by analyzing their context (i.e., neighboring words) in the corpus. It employs the distributional model, which states that similar words or phrases always appear in the same context in the corpus (Harris 1954). The output of a VSM is the vectors of words in an n-dimensional vector space where n corresponds to the number of unique words in the corpus. The semantic similarity between any two words is measured by employing the cosine angle between the vectors of the two words (Mao and Chu 2007).

Literature has reported different word embedding methods where the two widely used and computationally efficient word embedding methods include the word2vec and Glove (Muneeb et al. 2015). Word2vec employs artificial neural networks to learn the vector representation of words using the following two architectures, (1) continuous bag-of-words (CBOW), and (2) skip-gram (Mikolov et al. 2013). CBOW predicts a word by analyzing the vectors of a set of neighboring words, while the skip-gram predicts the neighboring words by considering the vector of a specific word. Both architectures produce comparable results; however, CBOW requires less time for the training process (Mikolov et al. 2013). The pretrained Glove model is based on a co-occurrence matrix which is produced by assuming that the probability of two words occurring together is equal to the dot product of their corresponding vectors (Pennington et al. 2014). Both word2vec and Glove are comparable in terms of performance; however, word2vec has better proved its effectiveness in performing NLP tasks (Si et al. 2019).

Related Studies and Research Gap

Several researchers have investigated the classification of legal documents into different categories to extract and manage legal text according to the needs. However, the current models may not support the classification of text into categories needed for design verification. Moreover, the current models are not applicable to prescriptive requirements where less semantic features are present within the requirement text that makes it challenging to identify the list of relevant risks addressed in requirement. In addition, the current models do not provide the ranking of labels in terms of relevance with the requirements to support the management of requirements in different styles or groups (using most relevant or less relevant requirements). For instance, Indukuri and Krishna (2010) proposed a classification framework to identify the payment-related clauses from e-contracts. The model employs the support vector machines (SVM) algorithms to classify e-contracts into payment related clauses and nonpayment related clauses. In another study, Glaser et al. (2018) compared the performance of six different machine learning (ML) algorithms to classify rental agreements into nine categories, including prohibition, permission, duty, etc. A few studies were focused on comparing the performance of deep neural networks with traditional ML algorithms. For example, recently, Wei et al. (2019) compared the performance of SVM with convolutional neural networks (CNN) to perform binary classification of four different legal datasets. The models developed in the studies discussed previously reached a performance ranging between 75% and 95%.

In the construction domain, literature has reported several studies focusing on the classification of contractual text. One of the initial studies in the text classification field was carried out by Caldas et al. (2002), who developed a classification model using five different ML algorithms to classify legal documents (i.e., specifications, change orders, and contracts) into 13 categories indicating the project components such as plumbing, schedule, structure, etc. Another study by Caldas and Soibelman (2003) implemented the same five ML algorithms to classify legal construction documents into 16 categories (concrete, mechanical, electrical, etc.). The hierarchical text classification technique was implemented in this study that achieved an higher accuracy of 95.33% in comparison with the 91.12% achieved by the flat classification technique implemented in the previous study by Caldas et al. (2002). A few studies also addressed the need of classifying the regulatory codes text to support automated compliance checking (ACC). In this regard, Zhou and El-Gohary (2016a) proposed the first ML-based model to classify the environmental regulatory codes into 10 different topics related to compliance checking (thermal insulation, air leakage, lightening power, fenestration, etc.). Following this study, Zhou and El-Gohary (2016b) implemented the ontology-based approach to compare the performance of current model with the previously developed ML-based model by Zhou and El-Gohary (2016a). The ontology-based model also performed the same task of classifying the environmental regulatory text into compliance checking topics.

The current models primarily employ the ML or neural network-based approaches that uses the semantic knowledge of the requirement text learned in the training phase to identify the requirement type in testing phase. However, the prescriptive requirements contain less or no semantic features in many cases. Therefore, in such cases, the developed models may not be able to identify the correct risk labels for requirements. In such scenarios, an unsupervised approach is more appropriate to identify the relevant requirements. Another limitation of the current model is the predefined number of labels and types that cannot be retrained on newly added labels of design verification. Therefore, a new model that can identify the relevant requirements from codes to perform design verification of bridges for different environmental risks is required. The aim of this paper is to provide an automated framework that can support the management of requirements into different clusters or groups, indicating different risk factors.

Proposed Framework

This paper presents an ontology-based framework that uses a proposed semantic knowledge base of risks and natural language processing to identify the risks addressed in design requirement texts specifically to support the management of design requirement in bridge engineering. The model employs NLP for measuring the semantic similarity between a design requirement text and the concepts of a risk ontology to predict the ranked list of relevant risk for the input requirement. This ranked list of relevant risks for each requirement can help bridge engineers in effectively classifying and organizing requirements according to risk factors.

Fig. 1 presents the methodology proposed in this study. First, a domain ontology modeling different types of risk to be controlled in the engineering design of bridges was developed by reviewing a text corpus including the design codes, state manuals, and published articles. Besides, a domain-specific vector space model was trained on the corpus using state-of-the-art word embeddings methods to convert the semantics of domain vocabulary into a computer-readable format. A similarity scoring method based on

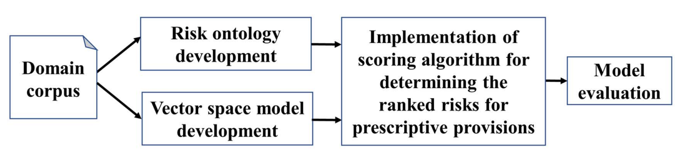

Fig. 1. Methodology of ontology-based requirement management system.

the vector space model was then proposed, which can measure and rank the relevant risk factors for a particular code provision. The risk factors addressed in the requirement text were ranked in a descending order where the risk factors with highest similarity score was most relevant and those with low similarity score were less relevant to the requirement. In this way, the model can assist users in identifying the more relevant and less relevant requirements for each risk factor to organize them for design verification. Finally, the performance of the model was evaluated on a set of labeled test requirements using different metric, namely Spearman, Kendall tau, and Pearson correlation coefficient.

Risk Ontology Development

A risk ontology, which is a core component of the proposed framework, was developed to represent the domain knowledge of the environmental risks associated with bridge engineering in a structured format. The ontology aims to classify different types of risk factors and model the semantic relationship between them. The present study adopted a well-established methodology to ontology development commonly used by previous studies (El-Gohary and El-Diraby 2010; Fidan et al. 2011; Zhou et al. 2016). The main steps involved in the development of the risk ontologies are described as follows.

Step 1: Development of Competency Questions

To develop the ontology, a list of competency questions that presents the scope of the risk ontology was created. The developed competency questions are as follows. What types of environmental risks are involved? In what scenarios are the risks present? What are the sources of risks? What are the consequences of risks? What is required to control the risks? What factors influence a specific type of risk? The aforementioned competency questions were used to formulate the ontological model, including concepts and relationships.

Step 2: Review of Existing Ontologies

This step includes a review of the existing ontologies relevant to the scope and competency questions discussed previously. An extensive literature search was conducted using various databases such as Web of Science, and Scopus. Since no relevant ontology on risks was found, a completely new ontology of environmental risks was developed for the present study.

Step 3: Development of a Risk Ontology

Prior to the development of the risk ontology, different types of environmental risks to be controlled in the bridge design were identified through an extensive content analysis of relevant domain-specific documents including the ASSHTO Bridge Design Specifications, the design manuals obtained from 50 State DOTs across the US and technical articles. Key concepts were identified and color-coded. The identified concepts were then classified and linked to each other using appropriate semantic relationships. Fig. 2 depicts the core part of the developed ontology. As shown in the figure, the present study identified the following nine types of risk: flood, earthquake, fire, snowfall, wind, temperature, overloading, vessel collision, and blast loading. These types of risk are organized in a three-level structure as the core ontology. The first level corresponds to the concept "risk factor," which is further classified into "natural risk" and "human-made risk." Please see Fig. 3 for more details of the classification. In this hierarchy, the relationship between a concept and its parent concept is modeled as "is-a."

Based on the core ontology, partial ontologies representing the knowledge associated with each of the nine risks were further developed using the same content analysis discussed earlier. The partial subontologies of two risk factors "temperature" and "flood" are illustrated in Figs. 3 and 4, respectively. The partial ontologies include the following four types of concepts: (1) subconcepts of each of the nine key risks, (2) concepts related to the structural elements of a bridge structure, (3) concepts related to structural responses to hazards, and (4) concepts about solutions to control structural

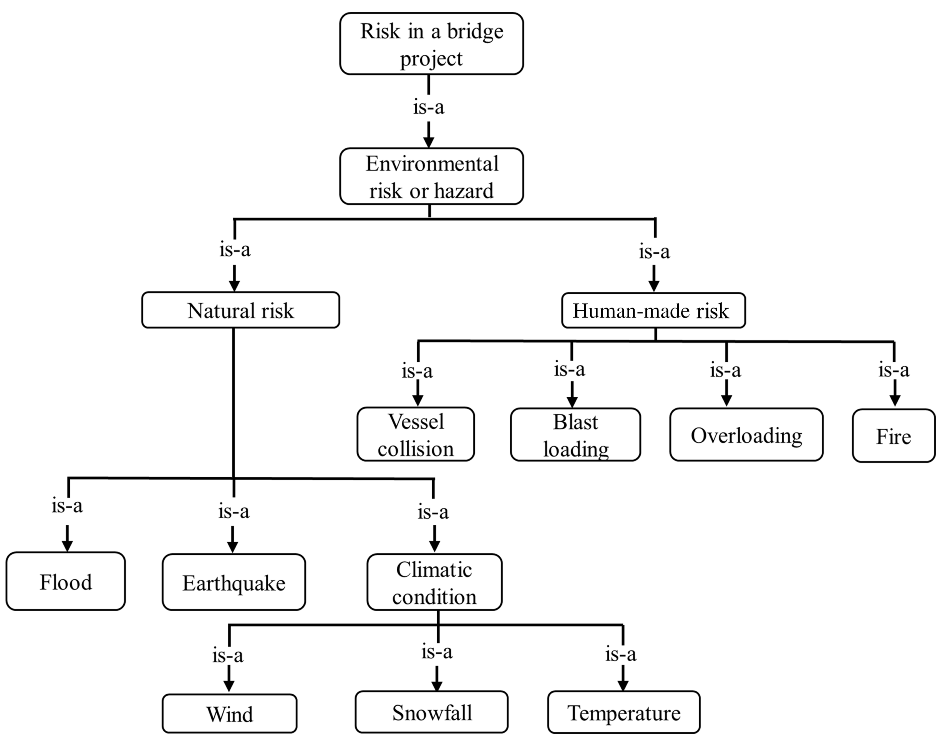

Fig. 2. Hierarchy of the environmental risks involved in bridge engineering.

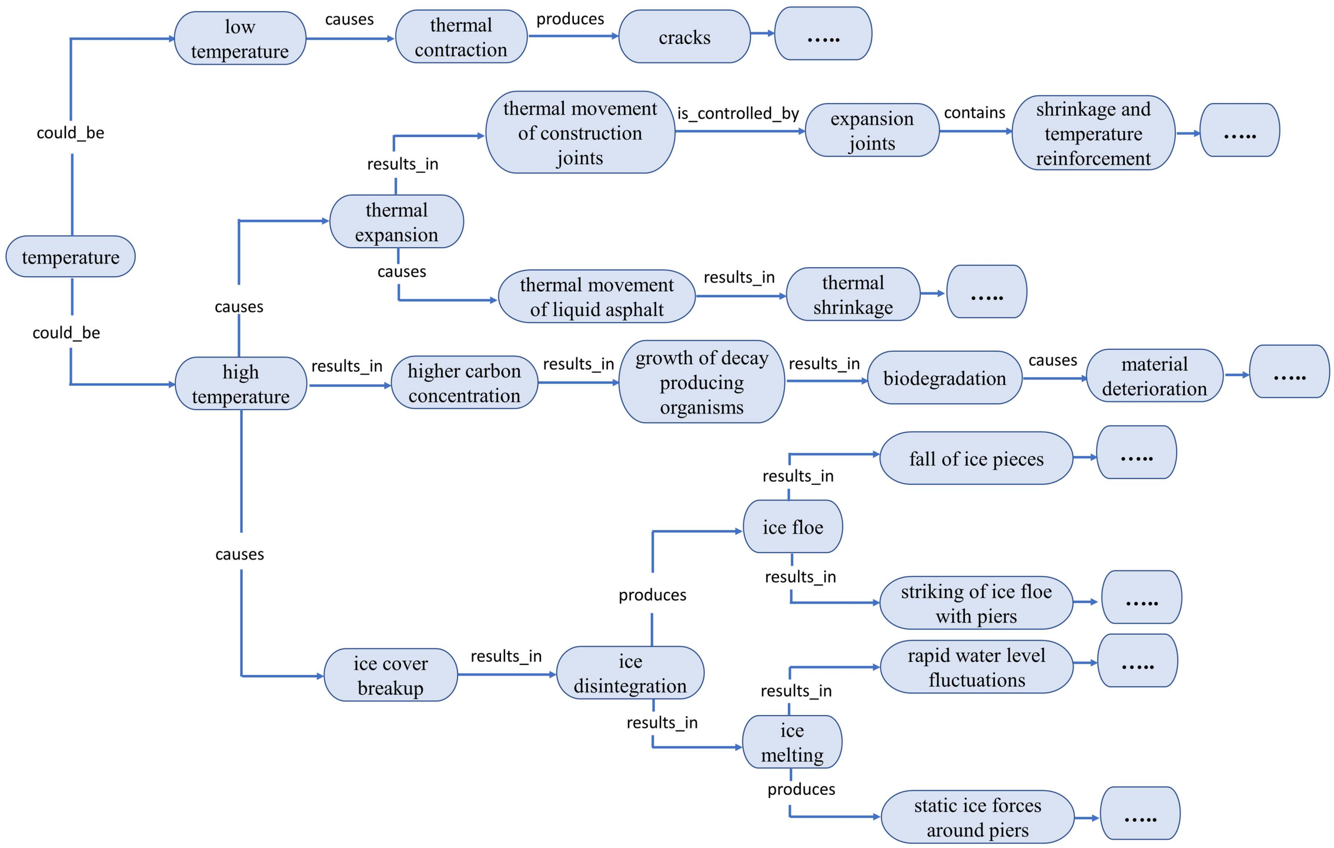

Fig. 3. Partial subontology for the risk category "temperature."

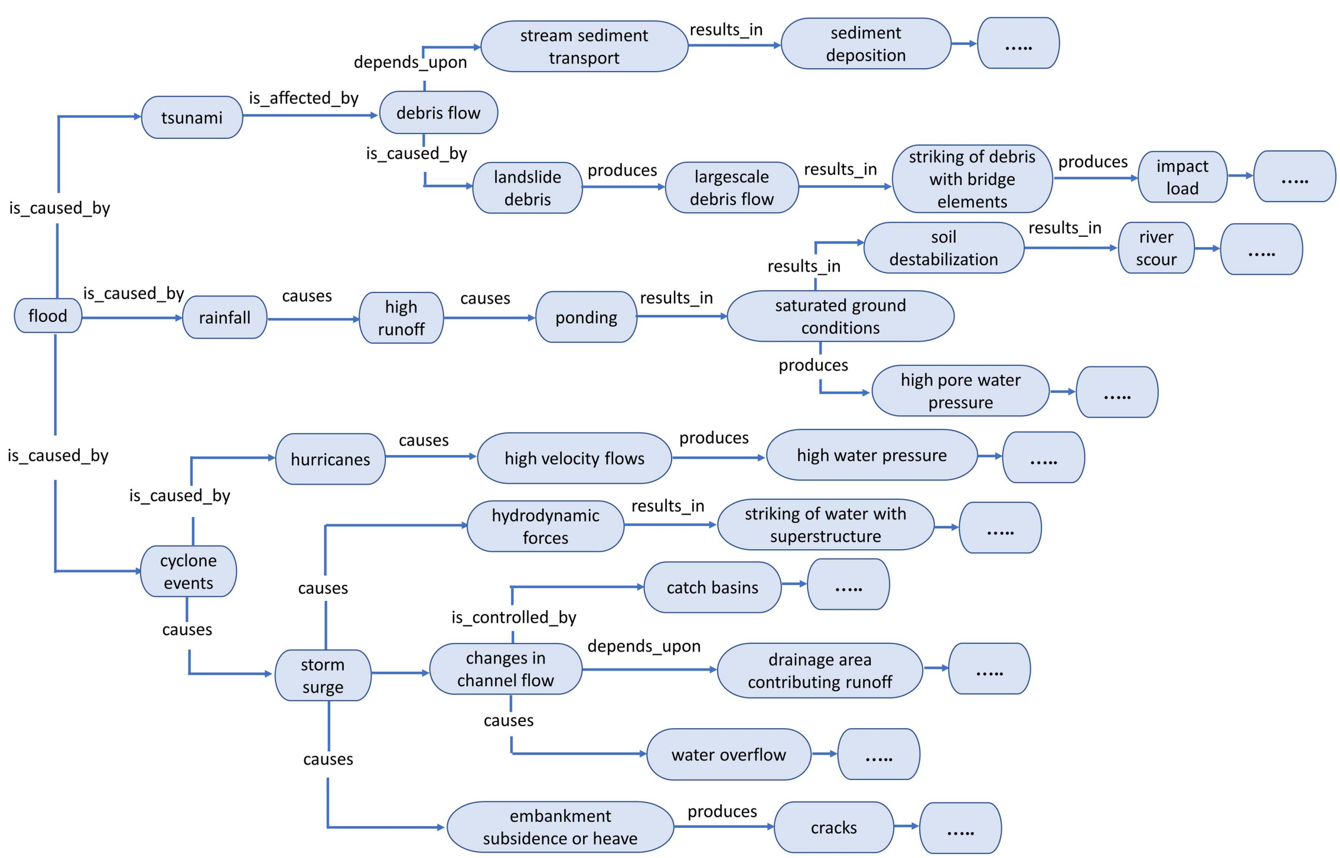

Fig. 4. Partial subontology for the risk category "flood."

responses to avoid structural failure. Let us take the partial ontology "temperature" (see Fig. 3) as an example. In that partial ontology, the risk factor "high temperature," which is a category under the "temperature" node, causes a structural response "thermal expansion." This response results in subsequent response "thermal displacement of the joints," which can be controlled by using "expansion joint." Note that a concept may occur in more than one partial ontology if it is semantically associated with more than one risk factor. For instance, "cracking" can result from an earthquake, wind, or flood, thus the concept "cracking" is found in the partial ontologies of these three risk factors. The developed risk ontology was evaluated (see Step 4) and finally coded using Protégé, an ontology modeling tool for use in the risk determination component of the proposed framework (see section "Scoring Algorithm for Determining the Relevant Ranked Risks of Code Provisions").

Step 4: Ontology Evaluation

This study employed the following two-stage procedure for ontology evaluation as suggested by El-Diraby et al. (2005). First, the content of the ontology was verified to ensure it provides complete answers to the competency questions. The authors found that the concepts are very relevant to the competency questions. Second, the ontology was evaluated by practitioners. We recruited a total of 10 experts with practical experience in bridge engineering. The background details of the experts are shown in Table 1. As shown, all experts were either structural engineers, designers, or managers who are involved in bridge designs or other structural designs. They have an average relevant experience in bridge designs ranging between 3 and 6 years. They were asked to complete a questionnaire regarding the completeness and preciseness of the concepts and their relationships. The survey includes six questions (see Table 2). The participants were asked to rate the ontology in terms of organization (i.e., ease of navigating the concepts) and classification of the concepts (i.e., preciseness and completeness of the concepts) using a 5-point Likert scale for rating the ontology, where 1 indicates "totally disagree" and 5 implies "totally agree." Table 2 presents the results of the survey regarding the experts' evaluation of the proposed ontology. As shown, the responses to most of the questions were positive. The participants strongly confirmed the ease of finding concepts, navigating through the ontology, and the preciseness of the classification. In addition, the participants indicated that they are familiar with the terms used in the ontology as an average score of 4.1 was given to Question 4. Furthermore, the results showed the participants' strong agreement with the relevance and coverage of the concepts (see Questions 5 and 6).

Table 1. Background details of the experts participated in ontology evaluation

No.	Position	Number of experts	Average experience in bridge designs (years)
1	Project developer and structural engineer	1	5
2	Structural engineer	3	4
3	Assistant project manager	2	4.5
4	Bridge engineer	1	3
5	Senior structural engineer	1	6
6	Project engineer	2	5

Vector Space Model Development

A vector space model is another crucial component of the proposed framework. As explained in the "Background" section, a vector space model is a collection of words in which each word is represented as a multidimensional vector. This model is necessary for measuring the semantic similarity between words, which is computed by the cosine similarity between the corresponding vectors. In this model, terms with similar semantics (i.e., synonyms) are close to each other. In order to develop such a model, an extensive corpus of texts related to bridge engineering was built using various bridge design manuals collected from the different state DOTs and the US DOT. Because the performance of the proposed approach highly depends on the vector space model, the data sufficiency requirements in terms of quality and quantity for the model training were carefully examined. The final corpus comprised a total of 1,016,049 words where the number of unique words was 27,845. NLP techniques were used to process this corpus and develop a vector space model. The subsequent subsections discuss the steps involved in the training of the vector space model in detail.

Text Preprocessing

Text preprocessing was applied to convert the domain corpus into a format for use by the vector space model. It typically excludes all the unnecessary features from the text corpus. First, the equations and the tables were removed from the corpus as they are not supported by the NLP techniques. Only plain text was included in the final corpus. The NLP techniques implemented for further preprocessing of text corpus are briefly discussed as follows:

- Lowercasing: Lowercasing was implemented to transform all corpus into a single lowercase format. It converts similar meaning words such as "Scour" and "scour" into one word "scour."
- Stop words and punctuation removal: English stop words that include is, are, am, the, and at were discarded in the text classification. These stop words are the words that appear with very high frequency in the corpus but carry low content and meaning. The punctuation was also removed before the training of the model.
- Tokenization: Tokenization was aimed at transforming a series of text into individual segments of words or tokens. Each token refers to a word, a numeric, or punctuation.

Table 2. Feedback of the experts and end-users on the ontology

No.	Questions	Evaluation feedback	Average rating	Median rating
1	Ease of finding concepts (a list of 10) in the risk ontology	"somewhat easy" to "easy"	3.7	3.8
2	Preciseness of the categorization of concepts in the risk ontology	"somewhat agree" to "agree"	3.6	3.8
3	Ease in navigation through the risk ontology	"somewhat easy" to "easy"	3.4	3.4
4	Familiarity with the terms used in the risk ontology	"familiar" to "strongly familiar"	4.1	4.2
5	Quality and relevancy of the concepts used in the risk ontology with bridge engineering design	"somewhat representative" to "representative"	3.9	4.2
6	Coverage of the main concepts of construction knowledge related to environmental risks	"covers" to "strongly covers"	4.4	4.2

- Lemmatization: Lemmatization returns the root term of each word in the corpus by stripping off the suffixes (Schütze et al. 2007). For instance, the words "accelerates," "accelerated," and "acceleration" were converted to a single root term "accelerate."

Training a Domain Vector Space Model

After preprocessing of the domain corpus, the vocabulary of the whole corpus was converted into a vector space where the semantics of each word was represented as a vector. The word2vec, which is an unsupervised method, was applied to develop the vector space model due to its high performance reported in the literature (Mikolov et al. 2013). In the word2vec model, the words are predicted given the context words in the corpus. The context words are controlled by selecting a window size. For instance, in the following requirement, "For the [overhead support structures the maximum deflection shall be limited to 2] inches per 100 feet of the span length," a window size of 5 is selected for the target word "deflection." The window size represented in the bracket shows the total number of words on the left and right side of the target word studied to comprehend the semantic meaning of the target word. The context of the word "deflection" will be {overhead, support, structures, the, maximum, shall, be, limited, to, 2}. In this study, both architectures of the word2vec algorithm including CBOW and skip-gram were applied separately to develop vector space models (Mikolov et al. 2013). CBOW predicts the target word based on the surrounding words, while the skip-gram algorithm predicts surrounding words considering the target word. As discussed, the corpus used for the training of the vector space model includes 1,016,049 words. Different values of the parameters involved in the training process were evaluated to select the best combination of values. The parameters included vector dimensionality (dimensions of word vectors), window size (number of neighboring words included in context window), and frequency threshold (minimum frequency for a word to be included in training process). The purpose of developing the vector space model was to learn the distributed representations and semantic similarities of the terms and concepts (Mikolov et al. 2013). Such a similarity score will be used to measure the relevancy of a concept in the risk category to the terminologies used in a specific prescriptive provision (Zhou and El-Gohary 2016a, b).

Fig. 5 shows the visualization of different terms of the developed vector space model. The skip-gram algorithm along with the frequency threshold, window size, and word vector dimensionality equal to 1, 15, and 900, respectively, was used. Principal component analysis (PCA) was applied to reduce the dimensions of vectors from 900 to 2 for plotting. The most related terms in the figure can be identified by observing the distance between the terms. For example, it can be seen that the terms as length, width, depth, and height, are close to each other as they commonly occur in the same context. A similar pattern can be seen for forces, loads, pressure, and stress.

Scoring Algorithm for Determining the Relevant Ranked Risks of Code Provisions

As mentioned earlier, the objective of this study is to develop a framework that enables requirement management for design verification by providing ranked list of relevant risk factors addressed in prescriptive code provisions. To achieve that goal, a semantic-based algorithm capable of recognizing the type of risk being controlled by a given provision was developed using the developed ontology and the vector space model. Fig. 6 illustrates the proposed algorithm. Given a certain input prescriptive provision, the algorithm will produce a ranked list of the most relevant types of risk

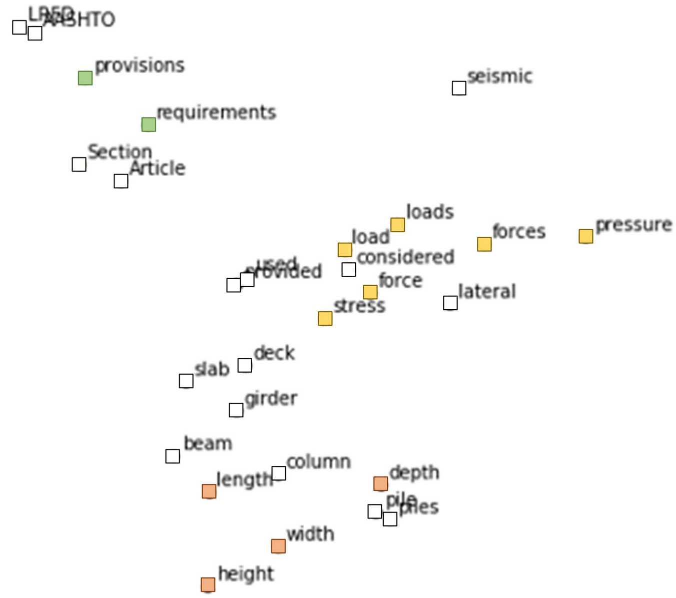

Fig. 5. Representation of word vectors of different terms extracted from the domain corpus.

based on the words used in the input provision. The algorithm includes two sequential stages.

Stage 1

This stage is to identify the concepts in the partial ontologies (see Figs. 4 and 5) that are highly relevant to each of the terminologies in the textual input provision. For that purpose, the algorithm calculates the semantic similarity between a term in the input provision and a concept (at a particular node) in the risk ontology using the pretrained vector space model. The similarity score s_{tc} between term t in the provision and concept c in the ontology is computed using Eq. (1), where V^t and V_r^c are the representation vector of term t and concept c, respectively. To eliminate irrelevant concepts, any similarity value s_{tc} lower than a predefined threshold is set to be 0

$$\text{Similarity score} = s_{tc} = \begin{cases} \dfrac{V^t \cdot V_r^c}{\|V^t\| \cdot \|V_r^c\|} \\ 0, \qquad\qquad if \ \text{it is less than } \theta \end{cases} \tag{1}$$

Stage 2

The second stage aims at measuring the relevance score between a concept and the whole input provision by totaling the similarity score s_{tc} from every term constituting the provision. Eq. (2) to compute the total similarity score S_c is described as follows:

$$\text{Total similarity score} = S_c = \sum_{t=1}^{T} s_{tc} \tag{2}$$

where T = number of terms in the input provision; and s_{tc} = similarity score between a term t with concept c.

Stage 3

This stage measures and rates the relevance between each of the nine risk factors (see Fig. 3) and the input prescriptive provision by summarizing the similarity S_c of every concept in the corresponding partial ontologies (see Figs. 4 and 5). The possibility of any potential biases due to the imbalance in the number of

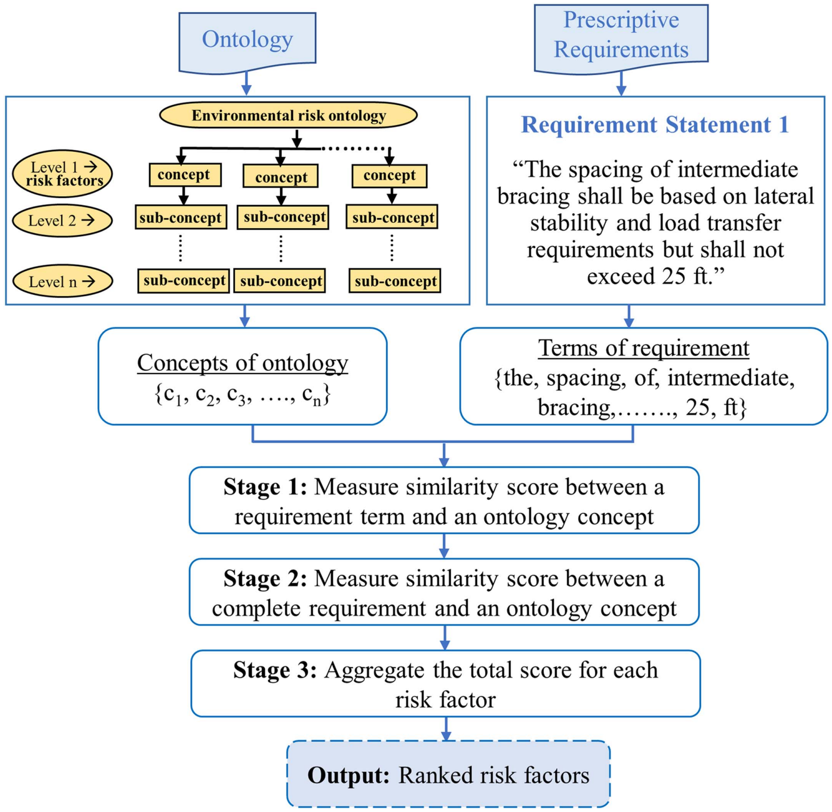

Fig. 6. Conceptual idea of the ontology-based ranking of the risks addressed in a prescriptive requirement.

concepts between the partial ontologies was minimized by applying the following two aggregation methods.

- Simple mean: This method measures the relevance score for a certain risk factor r by taking the average of the score of every concept in its partial ontology, as described in Eq. (3), where K is the total number of concepts in the partial ontology of the risk factor r, S_c is the similarity between of a concept c and the input provision, and n refers to the number of concepts with positive S_c

$$\text{Simple mean normalized score} = (S_r)_m = \frac{\sum_{k=1}^{K} (S_c)_k}{n} \quad (3)$$

- Weighted mean: This method assigns different weights to the similarity scores of different ontology levels while computing the mean. It can be argued that those items in higher levels of an ontology can provide more semantic abstraction than ones in lower levels. Thus, assigning greater weights to higher level concepts may help improve the identification of the risk factors. Eq. (4) was used to compute the weighted mean, where w_p indicates the weight assigned to those concepts at level p in the ontology, K_p is the total number of concepts in the partial ontology of the risk factor r at level p of ontology, S_c indicates the total similarity score of concept c, and n indicates the number of concepts with positive S_c

$$\text{Weighted mean score} = (S_r)_w = \frac{\sum_{p=1}^{P} w_p \times \left(\sum_{k=1}^{K^p} (S_c)_k\right)_p}{n} \quad (4)$$

Evaluation of the Proposed Risk-Based Requirement Management Method

Gold Standard and Evaluation Metrics

To evaluate the proposed method's capability in predicting the relevant risk factors addressed in design requirement, two gold standard datasets were developed including: (1) a set of 154 human-annotated design provisions extracted from the AASHTO Bridge Design Specifications; and (2) a set of 31 provisions randomly selected from different design manuals of US states. Each of the provisions was manually labeled with a ranked list of risk factors. The labeling was performed by two experts with essential domain knowledge in bridge engineering. Any disagreement arising in the labeling task was resolved by a third expert. For each test provision, its human-ranked risk types were compared with the automatically-ranked ones generated by the algorithm developed in Python. Three different rank correlation tests including Spearman, Kendall, and Pearson were applied to evaluate the performance [as illustrated in Eqs. (5)–(7)]

$$\text{Spearman coefficient} = r_s = 1 - \frac{6\sum_{i=1}^{n} (x_i - y_i)^2}{n^3 - n} \quad (5)$$

$$\text{Kendall tau coefficient} = r_k = \frac{2}{n(n-1)}(|C| - |D|) \quad (6)$$

$$\text{Pearson coefficient} = r_p = \frac{n\sum x_i y_i - \sum x_i y_i}{\sqrt{n\sum x_i^2 - (\sum x_i)^2}\sqrt{n\sum y_i^2 - (\sum y_i)^2}} \quad (7)$$

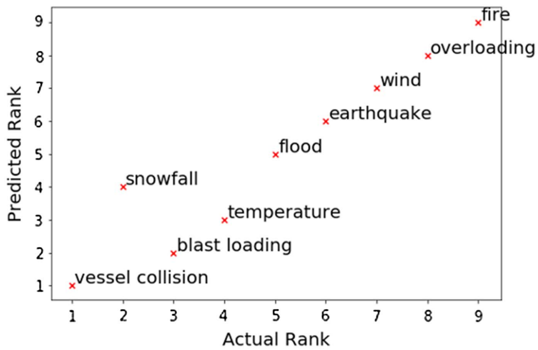

Fig. 7. Correlation plot of the test requirement. "The collision force and the point of application are assumed to vary linearly as the sound barrier setback varies between 30 and 122 cm (1 and 4 ft)."

where n = number of predefined risk categories; x_i and y_i = ranks of a risk category i by experts and the algorithm, respectively; C = a set of concordant pairs; and D = set of discordant pairs. These methods calculate the correlation coefficient, which indicates the level of agreement between human-annotated and predicted ranks of the risk labels for a given test provision. Ideally, the correlation coefficient will be equal to 1, indicating a complete agreement.

Computational Evaluation Results and Discussions

The proposed method and the calculation of the correlation coefficients were programmed in Python. Before developing the program, the ontology was coded in a digital format using the Protégé tool, and various vector space models were trained. These models serve as the primary knowledge sources to be used by the program to identify relevant risk factors. The program automatically generates a ranked list of risk factors associated with each instance in the AASHTO gold standard dataset. It then calculates the correlation between actual and predicted ranking. Fig. 7 presents an example of the correlation analysis for a challenging input provision. As shown, the six risk labels of vessel collision, flood, earthquake, wind, overloading, and fire were ranked accurately by the model. In comparison, the remaining three risk labels of snowfall, blast loading and temperature deviated from the actual ranks by a value of 1, 1, and 2 ranks, respectively. In an ideal case, all the points should form the linear line at an angle of 45°. Comparison between different vector space models that were trained using different word embedding methods and training parameters was conducted. Also, the parameters of the scoring method, including similarity thresholds and similarity score aggregation methods, were tuned to optimize the model performance. Analyses on the model performance across different types of risk were also conducted. The subsequent subsections discuss in detail the evaluation results when the model was tested on the AASHTO gold standard dataset.

Effects of Vector Space Models

Table 3 presents the results of the ranked risk prediction model using different pretrained vector space models. The CBOW and skip-gram algorithms were implemented to develop vector space models. In this initial analysis, the simple aggregation method was used for the total similarity scores computation. The total similarity scores of the nine risk factors for a specific requirement in the gold standard were computed by calculating average of the similarity scores of the concepts of a risk subontology and the requirement text. In addition, a similarity threshold θ was set to be 0 in the initial analysis which implies that all the similarity scores lower than zero were removed before computing the average scores. The threshold value was applied to ignore the impact of irrelevant text features. The ranks of risk factors were predicted according to their similarity scores with the requirement text. As shown in the table, the model prediction of ranks is well aligned with human judgment. Generally, the ideal value of 1 indicates a complete agreement of predictions with the human judgement. The correlation coefficient values produced by current models specifically by skip-gram models are also higher than 0.5 in many cases, which shows that the model is ranking the risks similar to an expert judgement. The highest Spearman, Kendall tau, and Pearson rank correlation coefficients are 0.7010, 0.5714, and 0.7008, respectively. These rank correlation results are the average of the 154 requirements in the gold standard. It can be seen that the model reliability is largely affected by the parameters of the vector space model. The optimal performance was achieved when the vector space model was

Table 3. Performance of the different vector space models (using simple aggregation methods and similarity threshold of 0)

			CBOW algorithm			Skip-gram algorithm		
Serial number	Window size	Vector dimensionality	Spearman correlation coefficient	Kendall tau correlation coefficient	Pearson correlation coefficient	Spearman correlation coefficient	Kendall tau correlation coefficient	Pearson correlation coefficient
1	2	300	0.1369	0.0983	0.1366	0.4894	0.3603	0.4891
2	5		−0.1837	−0.1333	−0.1837	0.5242	0.3928	0.5240
3	10		0.2495	0.1990	0.2495	0.6329	0.5072	0.6327
4	15		0.3124	0.2434	0.3125	0.6838	0.5515	0.6836
5	2	600	0.1517	0.1084	0.1513	0.4715	0.3495	0.4715
6	5		−0.1319	−0.0998	−0.1319	0.5363	0.4040	0.5360
7	10		0.2910	0.2278	0.2914	0.6281	0.5000	0.6279
8	15		0.3152	0.2452	0.3155	0.6983	0.5667	0.6980
9	2	900	0.1475	0.1059	0.1472	0.4861	0.3614	0.4861
10	5		−0.1355	−0.1012	−0.1356	0.5315	0.3993	0.5313
11	10		0.2976	0.2268	0.2973	0.6391	0.5112	0.6390
12	15		0.3256	**0.2387**	**0.3255**	**0.7010**	**0.5714**	**0.7008**
13	2	1,200	0.1524	0.1092	0.1522	0.4393	0.3224	0.4391
14	5		−0.0769	−0.0586	−0.0768	0.5407	0.4069	0.5405
15	10		0.0263	0.2044	0.2628	0.6382	0.5119	0.6380
16	15		0.2482	0.1968	0.2485	0.6932	0.5627	0.6931

Note: The bold values correspond to the highest coefficient values achieved that reflects the best performance.

trained using the skip-gram algorithm. It is also found that the performance of the model is significantly affected by the window size and the dimensions of word vectors. The model achieves the best performance when the window size and the dimensions of word vectors are set equal to 15 and 900, respectively. Between the two word embedding algorithms, the skip-gram algorithm significantly outperformed CBOW in all experiments. The highest coefficients when using the vector space model trained with the CBOW algorithm are only 0.3256, 0.2387, and 0.3255, respectively, for Spearman, Kendall tau, and Pearson.

Effects of the Scoring Method

In order to optimize the model performance, the algorithm was compared when varying the parameters of the scoring method. Accordingly, the semantic similarity threshold was changed from 0 to 0.45 with a step of 0.05. The authors also tested independently the two methods for aggregating similarity, including the simple mean and weighted mean as discussed earlier. For the weighted mean approach, the weight for those concepts in the first level was set equal to 1. The weight for children nodes is reduced by a step τ compared with its direct parent node. For example, if τ is set to be 10%, the weight for those concepts in the first, second, third, and fourth level of a partial ontology will be 1, 0.9, 0.8, and 0.7, respectively. In this experiment, we compared the models with $\tau = 10\%$ and $\tau = 25\%$.

Fig. 8 shows the model performance when varying the parameters of the scoring method. The results revealed that the correlation varied widely between different aggregating methods. The simple mean seemed to yield a significantly better result compared with the weighted mean. The highest Spearman, Kendall tau, and Pearson rank correlation coefficients of the model using simple mean aggregation are 0.7223, 0.6021, and 0.7222, respectively, at the similarity threshold of 0.15. The reason behind the poor performance of weighted mean is the considerable variation in the number of concepts at the same level among the partial ontologies. As a result, a risk factor of which the partial ontology includes many concepts in the higher levels always tends to receive a higher score. For instance, the requirement that "End anchorages (nuts washers and anchor plates) shall be compatible with the tie rod system and shall be galvanized in accordance with AASHTO M 111" is more related to the wind and overloading category as compared with the fire and temperature categories. However, the presence of more concepts in higher levels of fire and temperature risk categories produces higher ranks for them. In order to overcome this issue of favoring the risk factors having more concept at higher levels, the partial ontologies should have equal number of levels and a certain level of two ontologies should have equal concepts. In other words, the first level of partial ontologies of all nine risk factors should have equal number of concepts. However, this may not be possible since the nine risk factors require different number of concepts to explain their complete domain knowledge. The authors plan to try a few alternative averaging methods (weighted geometric mean, Bayesian average, etc.) in future work because geometric mean is very useful when the values in a series tend to make large fluctuations and when the values are dependent on each other. Bayesian average is also a potential alternative method since it considers a pre-existing belief to estimate the mean score of different risk categories having different number of values at a certain level. It covers the high variability in the number of results. However, since the weighted mean is a more popular method, it was implemented in this study. Regarding the similarity threshold, the optimal value is found to be 0.15. The correlation showed a slight increase when the threshold was varied from 0 to 0.15 but dramatically drop after exceeding the optimal point.

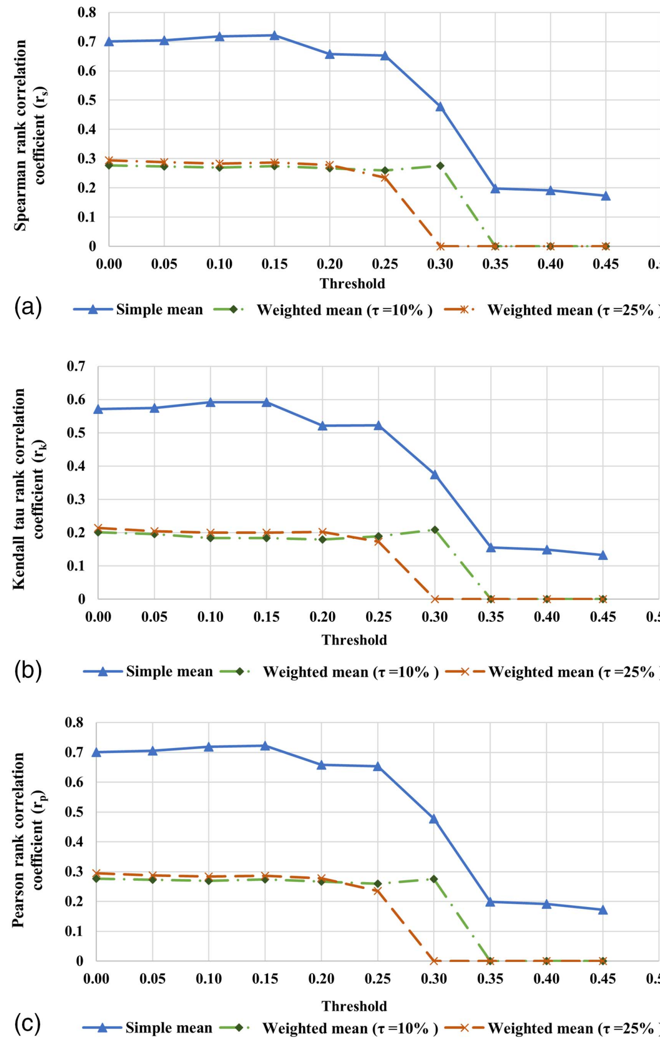

Fig. 8. Performance of the risk ranking model at different thresholds θ using different aggregation methods: (a) Spearman rank correlation coefficient at different thresholds; (b) Kendall tau correlation coefficient at different thresholds; and (c) Pearson rank correlation coefficient at different thresholds.

Model Performance across Risk Factors

For the analysis of performance variation between the risk factors, the optimal model found in the experiment discussed previously was used. Fig. 9 shows the mean deviation between the predicted rank and the actual rank for each risk factor. As shown, all the risk categories exhibited a low deviation ranging between 0.97 and 1.75. In other words, the proposed method offers considerable reliability in rating the relevance of a particular risk factor to an input provision. Moreover, the model performed best on "earthquake" and "fire," both of which yield a rank deviation of less than 1. Note that no risk categories produced a mean rank deviation of above 2 ranks. The highest mean rank deviation of 1.75 ranks was associated with the vessel collision risk factor.

Performance Comparison between AASHTO and State Gold Standard Datasets

Additionally, the developed model is potentially reusable to extract the required risk information from other types of federal design codes or manuals of any US state. The model can be used as-is for any bridge design document using the US terms and language.

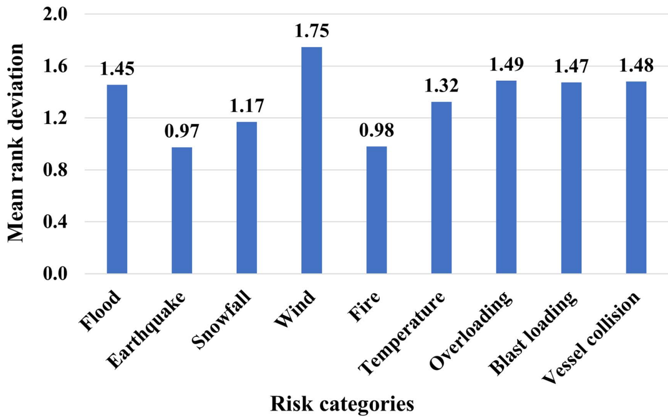

Fig. 9. Mean deviation of predicted ranks from actual ranks for different risk categories.

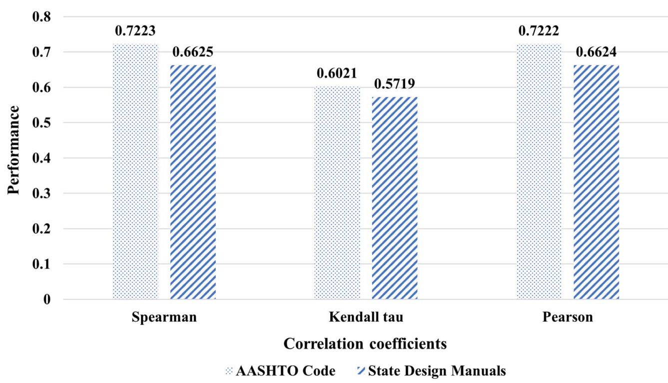

Fig. 10. Performance comparison of model on AASHTO code and state manuals.

However, in order to use the model for design codes of other countries, minor adaption is required in the ontology and vector space model according to the terms, language, and writing style in the new document. To evaluate the potential reusability of the current model on the state design manuals within the US, it was used without any modification on a dataset comprising provisions from state design manuals. The model was implemented to extract the information related to risk factors addressed in the randomly selected provisions. The performance of model on random sentences was tested against a manually labeled gold standard. The results of model on random sentences from design codes are shown in Fig. 10. As shown, the average Spearman, Kendall tau, and Pearson correlation coefficient of the model are 0.6625%, 0.5719%, and 0.6624%, respectively. This performance is comparable to the model performance achieved on AASHTO design codes, thus, implying the potential reusability of the developed model.

Practical Application of the Proposed Method: An Illustrative Example

The proposed framework was applied to a real-life practical design task performed by bridge engineers. Fig. 11 illustrates the specific tasks and steps involved in design phase where the model can be applied. The framework is specifically capable of supporting the task of identifying and grouping appropriate provisions for a certain project from state design manuals according to risk factors. Because design manuals are typically voluminous, manually reading lengthy documents to extract relevant provisions is tedious and error-prone. This illustrative example demonstrates how the model can assist engineers in quickly completing the task with a better outcome.

The US 701 bridge replacement project in South Carolina was selected for the example implementation for which the brief document was publicly available online. The project consists of the replacement of a 3.2 km (2 mi) section of US 701 over the Pee Dee River. Six young engineer practitioners were invited and accepted to participate in the study. The participants had an average experience of 1–2 years in bridge designs. They were randomly divided into control and experimental groups; each included three participants. They were provided with two design materials, including (1) a project brief specifying risk factors based on site investigations, and (2) a set of 117 unlabeled design provisions extracted from AASHTO code and South Carolina design manuals, including both relevant and irrelevant provisions (see Table 4). The participants were asked to read the design materials and map the provided

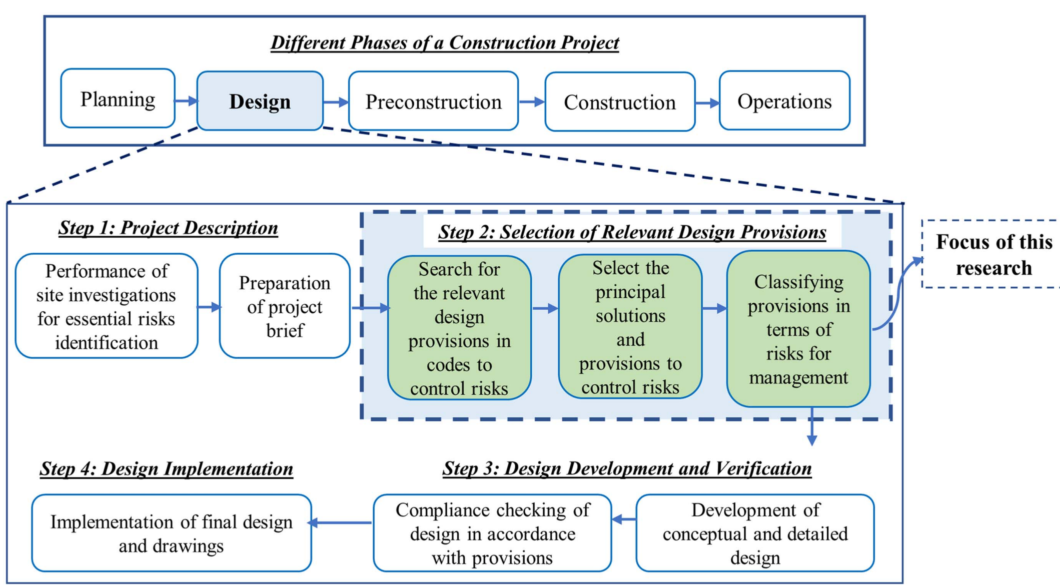

Fig. 11. Different phases involved in a construction project. (Design phase is the focus of this study.)

Table 4. Comparison of the performance of participants in the selection of the applicable code provisions to control risks

Risk category	Number of relevant requirements in the given requirements set	Accuracy of control group (%)[a]				Accuracy of experimental group (%)[a]			
		P1	P2	P3	Avg	P4	P5	P6	Avg
Flood	53	47	68	60	58	70	81	91	81
Earthquake	64	48	48	69	55	88	84	83	85
Snowfall	13	15	23	38	25	54	69	85	69
Wind	51	27	59	49	45	55	67	73	65
Fire	3	33	33	33	33	67	67	33	56
Temperature	21	24	38	24	29	62	57	71	63
Overloading	46	52	54	28	45	65	72	59	65
Blast loading	19	21	11	16	16	32	21	37	30
Vessel collision	37	43	57	68	56	89	84	78	84
Reading time	—	120	150	120	130	90	90	75	85

Note: P = participant; and Avg = average.
[a]The ranked risk information was not provided to the participants of the control group whereas the ranked risk information was provided to the participants of the experimental group.

provisions to the appropriate types of risk mentioned in the brief. The control group was asked to complete the task of using their own knowledge, while the experimental group was provided with the model's output of ranked list of relevant risk factors and engineering concepts. Fig. 12 illustrates the output of ranked risk factors for a given prescriptive provision presented to the experimental group. Each participant was given up to 2.5 h to read the project brief and to complete the task. The participants' performance was measured by comparing their results with the gold standard prepared by three practitioners having in-depth knowledge of bridge designs and verification. The performance by each participant was measured by the percentage of provisions correctly selected and reading time.

Results of Experimental Study

Table 4 shows the performance comparison between the participants in the control group and the experimental group. The results clearly show that the experimental group who used the model's output performed better than the control group who used the conventional method of performing the task in terms of both accuracy and reading time. As shown, the average time taken by the experimental group was about 35% less than the control group. With respect to relevant requirements extraction, the participants in the experimental group selected more accurate relevant provisions. The average accuracy of the control group across different risk factors ranges from 16% to 58%, whereas that measure of the experimental group varies from 30% to 85%. Overall, the findings showed at least 14% improvement when using the developed program. The best improvements were from snowfall and earthquake risk categories of which the average accuracy increased by 44% and 30%, respectively. Among the risk factors with the least changes were the blast loading (14%) and wind categories (20%). These findings are consistent with the comparison between the risk factors as discussed earlier in the section entitled "Model Performance Across the Risk Factors." This implies that the model is effective in the extraction of relevant applicable provisions from codes.

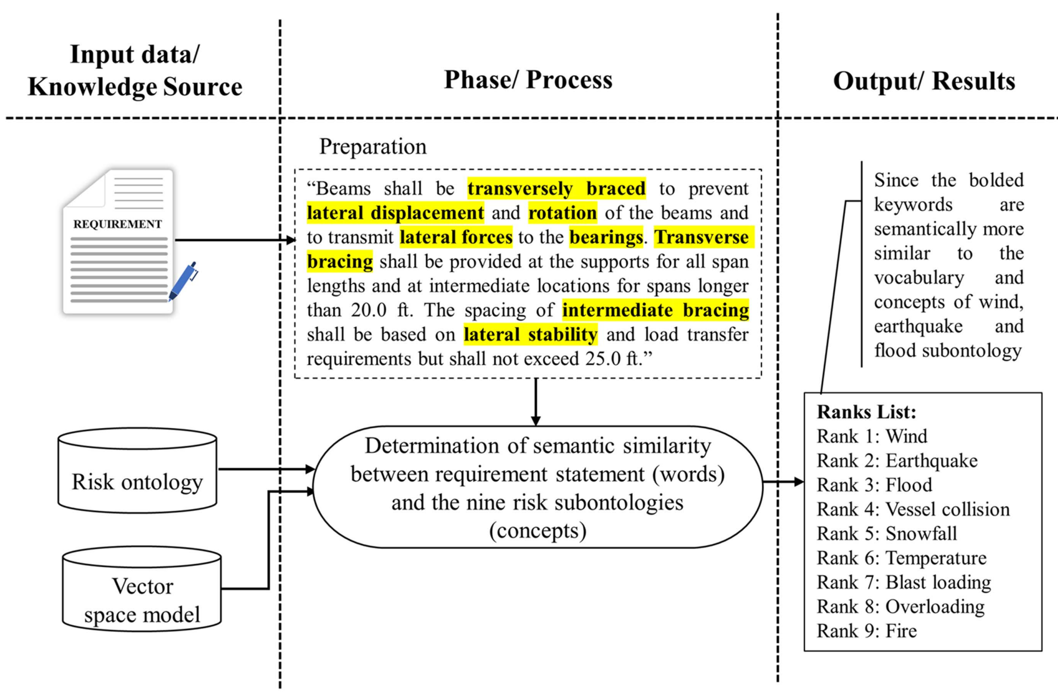

Fig. 12. Workflow of the proposed risk information extraction model.

Implications of the Study

In the realm of practice, the study provides several important implications for bridge designers. First, the study improves the comprehension and extraction of the requirements (prescriptive and performance-based requirements) from bridge design codes. Bridge designers with limited experience and domain knowledge often face difficulties in estimating the accurate intentions and risks addressed in requirements (especially the prescriptive requirements). This makes a challenging task for them to identify relevant risks for requirements to organize them according to risks and convert them into computer-readable rules. The developed model can reduce the human errors in extracting the precise set of requirements for each factor to check conformance of bridge designs to them. Second, the model can help in reducing time and effort required to estimate a cost-effective solution for a proposed project. It can predict the risks addressed in the requirement text, which can be used to compare the performance level of different prescriptive criteria provided in different standards. The study is an initial contribution to the development of system to predict risk types addressed in requirements of regulatory codes. The significance of the study from the industry perspective will be increased once the results are further improved. Currently, the ranked list of risk factors is the primary output of the model for every input requirement. Therefore, the user can see both input requirement text and the ranked list of risk factors in the output result. Consequently, this output information can be used for the organization and management of requirements in terms of risk factors. However, the model does not highlight which part of the input requirements is related to each of the risk factors. This is a limitation of the current model.

The study also contributes to the body of knowledge in many ways. For instance, a new ontology of the environmental risks involved in the bridge projects have been developed. In the construction domain, only a few attempts have been observed focusing on the utilization of domain knowledge to develop ontologies. The ontology developed in this study can be extended and modified to develop several other applications for the construction domain. Furthermore, the study is significant in a way because it is among the few studies in the construction domain that uses the semantic knowledge of the text through ontologies to develop automated frameworks rather than utilizing the supervised learning methods.

Conclusion

Bridge designs are performed according to a set of voluminous standards, regulations, and codes. The design verification of bridge designs requires the designer to extract and manage the applicable requirements associated with specific risk factors from codes to check the conformance of design to them. The requirements included in the standards and codes are largely prescriptive in nature where the unavailability of risks addressed in the requirements makes it challenging for a designer to extract the precise set of requirements relevant to a specific risk factor (earthquake, flood, etc.). The study has developed an automated risk-based requirement management model using the ontologies and deep learning methods. The developed system can rank the nine different types of risks addressed in the requirements according to their relevance and similarity with the requirement text. This ranking can help designers to extract the requirement relevant to a specific risk factor.

The core elements of the developed framework include a risk ontology, a vector space model, and a semantic similarity scoring method. The ontology, which covers nine types of risks typically controlled by bridge engineering, was developed through an extensive review of the design codes, standards, and published articles. The vector space model was trained on a corpus of 1,073,832 words comprising 27,845 unique words to convert domain vocabulary into numeric vectors. Two different algorithms including CBOW and skip-gram were implemented for the training of the vector space models with different values for the tuning parameters. The values of the vector dimensionality were tested ranging from 300 to 1,200 while the window size was tested ranging from 2 to 15. The experiments revealed that the skip-gram algorithm outperformed the CBOW algorithm at a frequency threshold, vector dimensionality, and window size of 1, 900, and 15, respectively. The skip-gram algorithm yielded an average Spearman, Kendall tau, and Pearson rank correlation coefficient of 0.7010, 0.5714, and 0.7008, respectively. Because the ontologies of nine risk factors contained different number of concepts, three different aggregation methods were evaluated for the computation of normalized total similarity scores. Among the methods, the simple mean averaging performed better than the weighted mean method. While calculating the normalized total similarity scores, the threshold values were also set where all the similarity values lower than the threshold were excluded before the computation of the normalized total similarity score. The best values of the average Spearman, Kendall tau, and Pearson correlation between the automated and manually ranked risk factors were reported to be 0.7223, 0.5920, and 0.7222, respectively, achieved at a threshold of 0.15.

The experimental results showed that the proposed approach and method's reliability in determining the risks hidden in the provisions is adequate. Still, the current study has certain limitations. The major limitation is the current performance of the model which is above 70% so far. The authors consider the current performance as good for the initial study; however, this performance needs to be enhanced before introducing the model in practice.

Data Availability Statement

All data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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